

RAHMA YASSER MAHMOUD

Data Analysis Instructor

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PROFILE

Data Analytics instructor and educator with a strong technical foundation in SQL, Power BI, and Data Warehousing, and an academic background in Bioinformatics (Top 3 of class, GPA 3.95/4). Combine ITI Data Visualization training with hands-on teaching experience currently a Teaching Assistant at the Faculty of Computers and Information, Assiut University, having led the Power BI module for FCI's Data Analysis Bootcamp. Passionate about translating complex technical concepts into clear, structured lessons, designing hands-on exercises, and mentoring students through real-world data and AI projects.

EDUCATION

Data Visualization Track

07/2025 – 10/2025

Information Technology Institute (ITI)

Assiut, Egypt

- Intensive professional track covering Power BI, Tableau, SQL, Data Warehousing (SSIS), and Excel.

Bachelor's degree in bioinformatics, Faculty of Computers and Information

09/2019 – 07/2023

Assiut University

Assiut, Egypt

- Ranked 3rd in class, GPA 3.95/4.00.
- Relevant coursework: Databases, Statistics, Machine Learning, Data Mining, Programming, Algorithms, Bioinformatics, R, Python.

EXPERIENCE

Power BI Instructor — Data Analysis Bootcamp

10/2025 – 01/2026

Faculty of Computers and Information (FCI), Assiut University

Assiut, Egypt

- Designed and delivered the Power BI module of an intensive Data Analysis Bootcamp targeting third- and fourth-year FCI students.
- Covered data modelling, Power Query, DAX measures and calculated columns, dashboard design, and data storytelling.
- Built hands-on lab exercises and real-world business case studies to reinforce each concept practically.
- Mentored participants on capstone dashboard projects, providing one-on-one feedback on data modelling, visual design, and presentation.

Teaching Assistant

03/2024 – Present

Faculty of Computers and Information, Assiut University

Assiut, Egypt

- Teach lab sessions and tutorials for undergraduate students across multiple core courses, including Data Mining, Artificial Intelligence, Machine Learning, Information Systems, and Databases.
- Prepare lab materials, hands-on exercises, and assessments aligned with course learning objectives.
- Provide one-on-one academic support during office hours and mentor students on graduation projects involving data analysis and machine learning.
- Grade assignments and exams and give written and verbal feedback to help students strengthen their technical and analytical skills.

COURSES TAUGHT

- **Power BI for Data Analysis** — FCI Data Analysis Bootcamp, 3rd & 4th year students.
Applied case studies translating raw data into interactive dashboards, from modelling through storytelling.
- **Data Mining** — Undergraduate, FCI, Assiut University.
Lab sessions covering data preprocessing, association rules, classification, clustering, and evaluation metrics.
- **Artificial Intelligence** — Undergraduate, FCI, Assiut University.
Tutorials and lab work on search algorithms, knowledge representation, and foundational AI techniques.
- **Machine Learning** — Undergraduate, FCI, Assiut University.
Hands-on labs on supervised and unsupervised learning, model training and evaluation, and applied ML projects in Python.
- **Information Systems** — Undergraduate, FCI, Assiut University.
Tutorials covering systems analysis and design, business processes, and the role of information systems in organizations.
- **Databases** — Undergraduate, FCI, Assiut University.
Lab sessions on relational database design, normalization, SQL, and practical query writing on real schemas.

TECHNICAL SKILLS & TOOLS

Databases & Data Warehousing: MySQL, MS SQL Server, Oracle (Toad), SQL / PL/SQL / Analytical SQL, SSIS.

Visualization & BI: Power BI (DAX, Power Query, modelling), Tableau, MS Excel.

Programming & Scripting: Python (Pandas, scikit-learn, Keras/TensorFlow), R, Bash, Git/GitHub, Linux.

Machine Learning & AI: Classical ML (Logistic Regression, Random Forest, XGBoost), Deep Learning, model evaluation.

Teaching & Communication: Lesson planning, lab design, mentoring, technical writing, presenting in English and Arabic.

LANGUAGES

Arabic (Native) English (C1 — comfortable teaching and presenting)

PROJECTS

Property Finder Real Estate Data Analysis & Visualization Project / *ITI Graduation Project* / [\(GitHub\)](#)

- Tools & Languages: Python | SQL Server | SSIS | Oracle SQL & PL/SQL | Power BI | Excel | Visual Studio.
- Built a complete BI system for real-estate analytics from large-scale public data scraping and cleaning to warehouse design and interactive Power BI dashboards. Used as a reference example when explaining the full BI lifecycle to students.

Gravity Books Data Warehouse Project / [\(GitHub\)](#)

- Tools & Languages: SQL Server | SSIS | T-SQL | Visual Studio.
- Converted a normalized transactional database into a dimensional data warehouse designed fact tables (Fact_Orders, Fact_Orders_Acc) and supporting dimensions (Book, Author, Customer, Shipping Method, Order Status, Date) to enable analytical sales reporting.

Customer Churn Rate Analysis Dashboard / [\(GitHub\)](#)

- Tools & Languages: Power BI.
- Power BI dashboard analyzing customer behavior and churn patterns to support data-driven decisions; reused as a teaching case for DAX measures and storytelling.

Road Accident Data Analysis Dashboard (Excel) / [\(GitHub\)](#)

- Tools & Languages: Microsoft Excel (Pivot Tables, Pivot Charts, Slicers, Calculated Fields).
- Built an interactive dashboard analyzing 417,883 road-accident casualties by severity, vehicle type, road and weather conditions, and location, with year and urban/rural slicers demonstrates advanced BI dashboarding using Excel alone.

Sales Performance Analysis for Electronics Retail Dataset / [\(GitHub\)](#)

- Tools & Languages: Analytical SQL (Aggregations, Window Functions, Ranking, Pivoting).
- Evaluated sales performance across regions, product categories, and salespersons using analytical SQL; reused as a lab exercise for SQL students.

DBMS Bash Scripting Project / [\(GitHub\)](#)

- Tools & Languages: Bash, Linux.
- Built a file-based Database Management System entirely in Bash implemented full CRUD operations (create, insert, select, update, delete, drop tables) through modular shell scripts and a menu-driven CLI, with no external database engine.

Glaucoma Prediction using Deep Learning / *Bachelor Graduation Project* / [\(GitHub\)](#)

- Tools & Languages: Django, PostgreSQL, Python, Keras, TensorFlow, HTML, CSS, JS.
- Built a website to help ophthalmologists predict glaucoma from fundus images, reaching ~90% validation accuracy.

LUAD Gene Prioritization Project / [\(GitHub\)](#)

- Modular Python pipeline integrating TCGA tumor RNA-seq and GTEx normal tissue data; applies a semi-supervised positive-unlabeled (PU) learning approach with classical ML models (Logistic Regression, Random Forest, XGBoost) to prioritize candidate genes from real biomedical data.